K1391 — Casey's two questions (scale + 137) are one honesty issue seen twice, and the corpus reconnect found the ghost hiding in our OWN ledger. ★ Q1 (Planck scale): YES — taking the Planck mass as the ONE dimensionful input is legitimate and GR-level (GR takes G; every theory takes one scale). This reframes a_4/gravity from "scale hole = can't derive SM" to "derives the SM up to one dimensionful anchor, stated plainly like GR states G" — correcting a subtle UNDER-claim. BUT the 8π is a DIMENSIONLESS factor (which Planck mass — standard vs reduced), not a scale; it still must be derived forward, and taking it as input would be a fudge. ★ Q2 (137): right to push — don't gate, investigate — AND the key reconnect is that forward-137 = the boundary channel-capacity = the SAME RS/7-bit-layer object we derived all weekend. But FIRST: the corpus contradicts itself — **`BST_ElectronMass_Derivation.md`** line 404 labels the Wyler α formula "Proved (in BST)" while K676/K680 RETIRED it (Identified, Robertson 4-reading). That stale "Proved" IS the ghost — reconcile it before any Connes approach; we cannot defend 137 while our own ledger calls the retired Wyler formula proved. Plus ratify the team's honest closes + Lyra's B1 finiteness gift + the C4 idempotent base-camp. Cal cold-reads.

Keeper (2026-08-11, ~13:25 / team clock ~16:25, near EOD. Casey asked why we can't just take the Planck scale, and pushed to derive/defend 137 before Connes. Both are right instincts; both need one precision. And the reconnect found a stale over-claim in our own corpus that's exactly the ghost. Cal cold-reads. Nothing pushed.)

★ Q1 — the Planck scale IS a legitimate single input (GR-level), but separate the scale from the 8π

Casey: *"Why can't we introduce the Planck scale (mass) to derive the SM? Like BST, we need units/scale for comparison."* Correct, and it fixes an under-claim we've been making: - Every theory takes exactly one dimensionful input. GR takes G (or equivalently M_Planck); the SM takes the Higgs VEV / Λ_QCD; BST takes ℓ_B = the Planck length anchor (banked, CLAUDE.md). Taking the Planck mass as the ONE scale is not a fudge — it is what a fundamental theory is *allowed* to do. So a_4's "scale hole" is mis-framed: the *dimensionful* part is a legitimate single input. a_4 derives the SM bosonic Lagrangian up to one dimensionful anchor — which is exactly GR's honesty (GR predicts everything up to G). State it plainly, don't apologize (feedback: don't-diminish-Identified, state-inputs-like-GR). - BUT — and this is the precision — the 8π is NOT a scale. The m_e = $6\pi^5\cdot\alpha$^($2C_2$)·M_Planck anchor's open question is *which* Planck mass — standard M_Planck (n=12=$2C_2$) vs reduced M_Planck/$\sqrt{(8\pi)}$ (n=11.67). That factor of 8π is dimensionless — a pure number, not a unit. You cannot take a dimensionless factor as "input"; that would be exactly the fudge. So: the *dimensionful anchor* is a legitimate GR-level input (resolves the apologetic "scale hole" framing); the *dimensionless 8π* remains the forward make-or-break (this weekend's cell-count/Clifford work). Reframe helps the honest framing without dissolving the 8π. Calibrate both directions: don't under-claim (the scale IS a legit input) and don't over-claim (the 8π still must be derived).

★ Q2 — the 137 push: right, reconnected, but reconcile our OWN ledger first

Casey: *"Everybody is scared of 137; we keep it to the appendix but we have a rich corpus and should be able to derive/defend it before Finster and Connes."* Agreed — investigate, don't gate. Two findings: 1. ★ THE RECONNECT (this is the exciting part): forward-137 = the boundary channel-capacity = the SAME object as this weekend's summit work. The corpus mechanism for α^{-1}=137 is "boundary channel-CAPACITY / charge-count on the conformal boundary" — and all weekend we established the boundary as a *channel* (RS-coding on GF(128), the 7-bit layer, 137 living in that layer). So 137 is not a separate appendix item to defend — it is the channel-capacity of the very boundary whose bit-count gives the gravity/DM 16. Deriving 137 forward and deriving the cell-count are the same program: *what is the exact channel-capacity of the D_IV^5 boundary?* If that count is forced to 137, it derives α AND grounds the 7-bit RS layer. This is a real lead, not a defense. 2. ★ THE GHOST IS IN OUR OWN LEDGER (reconcile before Connes). `BST_ElectronMass_Derivation.md` line 404 carries: *"$\alpha = (9/8\pi^4)(\pi^5/1920)$^{1/4} = 1/137.036 | Proved (in BST) | Wyler formula on D_IV^5."* But the constants file + K676/K680 RETIRED the Wyler integral (Robertson trap, 4-reading) — α is IDENTIFIED, not Proved. This stale "Proved" label is precisely Wyler's ghost, sitting in our repo. If Connes (or a grad student) greps our corpus and finds the retired Wyler formula labeled "Proved," that is the dismissal handed to us. RULING: the line-404 "Proved" must be reconciled to "Identified / Wyler-retired (K676/K680)" — a straggler fix, Grace's data-lane, Keeper-ruled — BEFORE any external 137 defense. You cannot defend a number your own ledger both proves and retires. 3. The forward-derivation bar (to move 137 out of the appendix, honestly): the channel-capacity count must be forced to exactly 137, blind — with the decoys refused: 137 = N_c^3·n_C+rank (the forward form) AND 128+9 (=2^g+N_c^2) AND the Wyler integral are THREE provenances, and a number with three faces hasn't told you which. Force one, blind; refuse the other two. If the boundary channel-capacity derivation (Q2.1) forces 137 without reference to the decimal, 137 earns the main text. Until then: appendix, Identified.

Ratify the team's honest closes + real advances

- Grace's tiers (exactly right): T2555 (boundary stores polarity) = Derived-core (totally-real half-dim + SO(2)-phase-drop are theorems) + Structural caveat (one-bit-per-direction owed); T2556 (16&4 = two faces of Cl(4)) = Structural (16=4^2 rigorous; "storage = full endomorphism algebra" owed); T2557 (same-object) = Conditional on n_C−1=2·rank (holds, but both 16/3's stay PD till the forcing step). Clean, honest, no dressing.
- The DESI correction propagated to the WHOLE team — Cal, Elie, and Lyra each independently owned the ~3σ over-statement and converged on ~1.3–1.6σ + banked-w=−1-safe + Σm_ν-is-the-real-exposure. My K1389 self-correction was absorbed cleanly across all four CIs within one round — the walk-back rate stays healthy (K1383), including on the auditor's own error. This is the discipline at its best: a correction that started with the auditor admitting he over-claimed, adopted by everyone without defense.
- ★ Lyra's B1 "gift" (real advance): the non-compactness is only the *symmetry group* SO(5,2); the *domain* is a bounded ball with finite volume → Finster's causal action is finite for free. This resolves one of B1's three owes (finiteness on the non-compact domain). B1 now owes TWO, not three: (a) the Bergman projector IS Finster's causal-action projector (same object), (b) second-variation → minimum not just critical point. Cleaner target.
- Lyra's C4 "honest half" — real but it's base-camp, not half the summit (calibrate): for an idempotent projector, $\int\int$P(x,y)P(y,x) = Tr(P^2) = Tr(P) *exactly* — so Finster's *unweighted* action = Connes' trace, a genuine exact identity. Ratify it as a real foundation. BUT hold the calibration: Finster's causal content IS the weighting (the spectral |P|^2-weighting is what makes it "causal"); the unweighted idempotent identity is the *trivial* part, true for any projector. So C4 has base-camp established (the idempotent identity), and the summit — the weighted equality = Connes' f — is 100% of the physics and unstarted. Do NOT tier "C4 half-proven"; tier "C4 base-camp (Structural) + summit (scaffold, the weighting)." The mountain is the weighting.

Route (final hour / near EOD)

1. ★ 137 reconnect (Casey's push, real lead): Lyra + Grace — derive the boundary channel-capacity forward (= the RS 7-bit-layer count); is it forced to 137, blind? Refuse 128+9 and the Wyler integral. First: reconcile the line-404 "Proved" → Identified (Grace data-lane).
2. ★ B1 (now 2 owes): Lyra — the same-object projector identity + second variation (finiteness banked via the bounded-ball gift). Cal gates Palais.
3. C4: the idempotent base-camp is banked; the summit is the *weighting* → Connes' f. Write the weighted operator identity or name the gap. Scaffold.

4. a_4: Derived-structure form banked (Elie); the scale is a legit GR-level input, the 8π dimensionless factor is the open piece (route to the cell-count).
5. DESI: Elie's one light-cone computation, Cal step-audited; then park (DR3). Reconciled to 1.3–1.6σ.
6. Keeper — recused from derivations; hold the tiers, the scale-vs-8π precision, the 137 anti-numerology bar, and the ghost-reconciliation gate. Near EOD — this is a clean checkpoint.

— Keeper, K1391, 2026-08-11. Q1 (Planck scale): taking M_Planck as the ONE dimensionful input is legitimate + GR-level (GR takes G) — reframes a_4/gravity from "scale hole" to "derives SM up to one anchor, stated like GR states G" (fixes an under-claim); BUT the 8π is DIMENSIONLESS (which Planck mass), not a scale — still owed forward, taking it as input = fudge. Q2 (137): investigate not gate; RECONNECT: forward-137 = boundary channel-capacity = the SAME RS/7-bit-layer object as this weekend (not a separate appendix defense — the channel-capacity of the boundary whose bit-count gives 16). GHOST IN OUR LEDGER: BST_ElectronMass_Derivation line 404 labels the Wyler α formula "Proved" while K676/K680 RETIRED it (Identified) — reconcile to Identified BEFORE any Connes approach (Grace data-lane, Keeper-ruled); can't defend a number our own ledger both proves + retires. Forward bar: channel-capacity forced to 137 BLIND, refuse the 3 provenances ($N\_c^3 \cdot n\_C$+rank, 128+9=2^g+N_c^2, Wyler integral). Ratify: Grace tiers (T2555 Derived-core+Structural-caveat, T2556 Structural, T2557 Conditional); DESI correction propagated to ALL FOUR CIs cleanly (walk-back healthy incl. auditor's own error, K1389/K1383); Lyra B1 gift (bounded-ball finite volume → causal action finite for free, resolves 1 of 3 owes, now 2: same-object projector + 2nd-variation); Lyra C4 idempotent identity (unweighted Finster = Connes trace, EXACT) = real base-camp BUT the trivial part — the WEIGHTING is Finster's causal content = 100% of physics = the summit, scaffold; tier base-camp Structural + summit scaffold, NOT "half-proven." Route: 137 forward channel-capacity (reconcile line-404 first); B1 2 owes; C4 weighted summit; a_4 scale=GR-input/8π-owed; DESI Elie one computation Cal-audited then park; Keeper holds tiers+precision+ghost-gate. Near EOD, clean checkpoint. Cal cold-reads. Nothing pushed.